



Intent

- An Android Intent is an abstract description of an operation to be performed.
- It can be used with **startActivity** to launch an Activity

Below are some applications of Intents:

- Sending the User to Another App
- Getting a Result from an Activity
- Allowing Other Apps to Start Your Activity



There are two types of intents in android

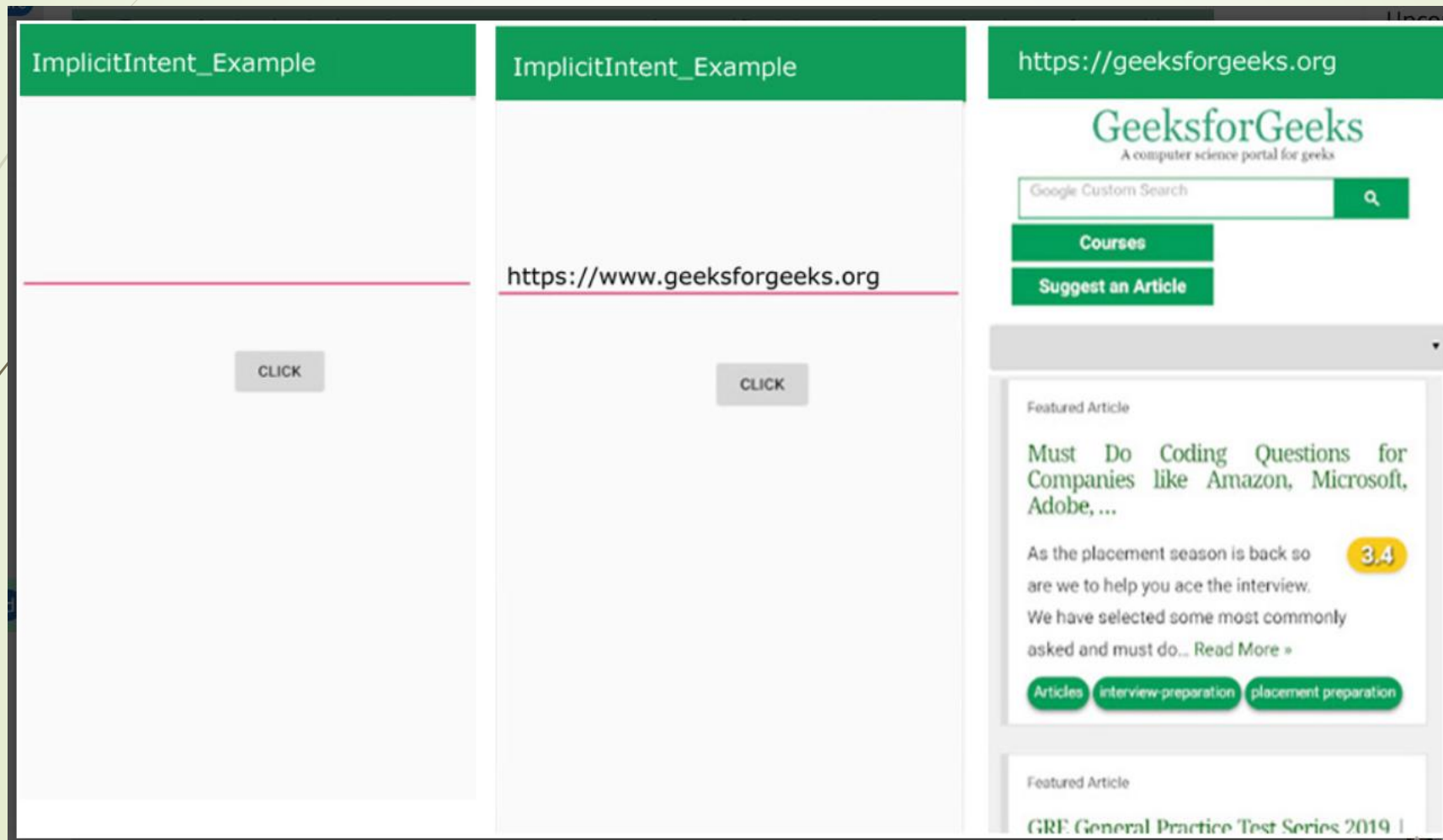
- Implicit

- Explicit

Implicit Intents

```
Intent intent=new Intent(Intent.ACTION_VIEW);  
  
intent.setData(Uri.parse("https://www.geeksforgeeks.org/"));  
  
startActivity(intent);
```

For Example: In the below images, no component is specified, instead, an action is performed i.e. a webpage is going to be opened. As you type the name of your desired webpage and click on the 'CLICK' button. Your webpage is opened.



Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="fill_parent"
```

```
    android:layout_height="fill_parent"
```

```
    android:orientation="vertical">
```

```
    <TextView
```

```
        android:id="@+id/textView"
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:text="Enter the Website name"
```

```
    />
```

```
<EditText
```

```
    android:id="@+id/editText"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:ems="10"
```

```
    android:inputType="textPersonName"
```

```
    android:text=""/>
```

```
<Button
```

```
    android:id="@+id/button"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:text="Visit"/>
```

```
</LinearLayout>
```

MainActivity.java

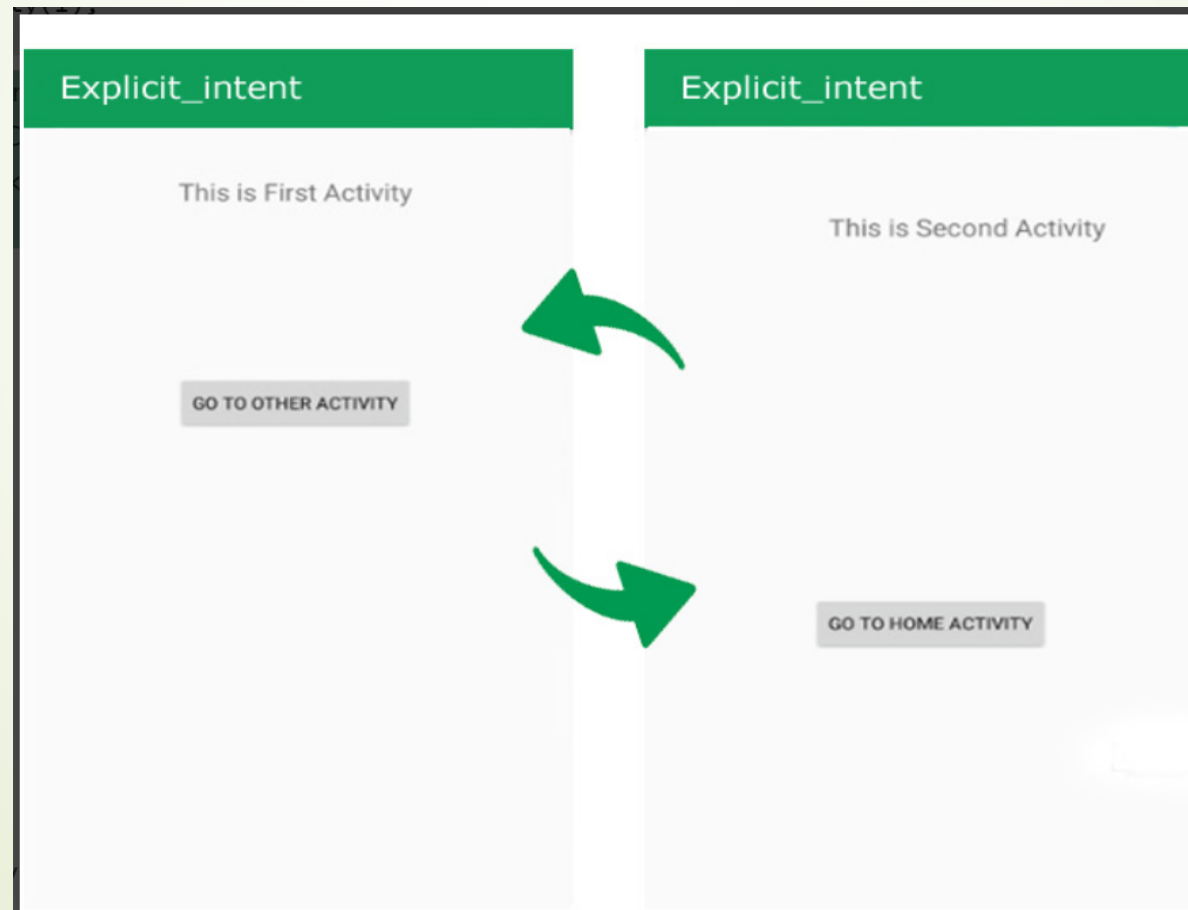
```
public class MainActivity extends AppCompatActivity {  
  
    Button button;    EditText editText;  
  
    @Override  
    protected void onCreate(Bundle savedInstanceState) {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
  
        button = findViewById(R.id.button);  
        editText = findViewById(R.id.editText);  
        button.setOnClickListener(new View.OnClickListener()  
        {  
            @Override  
            public void onClick(View view) {  
                String url=editText.getText().toString();  
                Intent intent=new Intent(Intent.ACTION_VIEW, Uri.parse(url));  
                startActivity(intent);  
            }  
        });    }  
}
```

Explicit Intents

```
Intent i = new Intent(getApplicationContext(), ActivityTwo.class);  
  
startActivity(i);
```



For Example: In the below example, there are two activities (FirstActivity, and SecondActivity). When you click on the 'GO TO OTHER ACTIVITY' Button in the FirstActivity, then you move to the SecondActivity. When you click on the 'GO TO HOME ACTIVITY' button in the SecondActivity, then you move to the first activity. This is getting done through Explicit Intent..



Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<LinearLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="fill_parent"
```

```
    android:layout_height="fill_parent"
```

```
    android:orientation="vertical">
```

```
    <TextView
```

```
        android:id="@+id/textView"
```

```
        android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
```

```
        android:text="First Activity"
```

```
    />
```

```
<Button
```

```
    android:id="@+id/button"
```

```
    android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
```

```
    android:onClick="onClick"
```

```
    android:text="Click here to go to another  
activity"/>
```

```
</LinearLayout>
```

MainActivity.java

```
public class MainActivity extends AppCompatActivity {  
    Button button1;  
  
    @Override  
    public void onCreate(Bundle savedInstanceState)  
    {  
        super.onCreate(savedInstanceState);  
        setContentView(R.layout.activity_main);  
  
        // Bind the components to their respective objects  
        // by assigning their IDs  
        // with the help of findViewById() method  
        button1 = (Button)findViewById(R.id.button);  
        button1.setOnClickListener(new OnClickListener() {
```

```
        public void onClick(View view)  
        {  
            // Creating explicit intent  
            Intent i = new Intent(getApplicationContext(),  
                                    ActivityTwo.class);  
            startActivity(i);  
        }  
    });  
}
```



New activity can be created as

- file -> new -> activity -> empty activity

Activity_two.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<FrameLayout
```

```
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
android:layout_width="fill_parent"
```

```
android:layout_height="fill_parent"
```

```
android:orientation="vertical">
```

```
<TextView
```

```
android:id="@+id/textView11"
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:text="Second Activity"
```

```
android:layout_gravity="bottom"
```

```
/>
```

```
<ImageView
```

```
android:src="@drawable/logo"
```

```
android:scaleType="fitCenter"
```

```
android:layout_width="fill_parent"
```

```
android:layout_height="fill_parent"
```

```
/>
```

```
<Button
```

```
android:id="@+id/button11"
```

```
android:layout_width="wrap_content"
```

```
android:layout_height="wrap_content"
```

```
android:onClick="onClick"
```

```
android:text="Click here to go to First  
activity"/>
```

```
</FrameLayout>
```

ActivityTwo.java

```
public class ActivityTwo extends AppCompatActivity {
```

```
    Button button1;
```

```
    @Override
```

```
    public void onCreate(Bundle bundle)
```

```
    {
```

```
        super.onCreate(bundle);
```

```
        setContentView(R.layout.activity_two);
```

```
        // Bind the components to their respective objects
```

```
        // by assigning their IDs
```

```
        // with the help of findViewById() method
```

```
        button1 = (Button)findViewById(R.id.button11);
```

```
        button1.setOnClickListener(new OnClickListener() {
```

```
            public void onClick(View view)
```

```
            {
```

```
                // Creating explicit intent
```

```
                Intent i = new Intent(getApplicationContext(),
```

```
                    MainActivity.class);
```

```
                startActivity(i);
```

```
            }
```

```
        });
```

```
    }
```

```
}
```